



Computational fluid dynamics simulations of fluid flow through the packed beds: A study of contact point treatments

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ABSTRACT

This study presents a comprehensive computational investigation of the effects of inter-particle contact point treatments on pressure drop predictions in fluid flow through fixed beds. Using Reynolds-averaged Navier–Stokes simulations, the impact of particle diameter adjustments ($\pm 0.02\%$ to $\pm 10\%$) on flow characteristics, mesh quality, and numerical accuracy was systematically evaluated. A comparison of gap and overlap methods for a packed bed filled with spherical particles of diameter D_p demonstrates that minor geometric modifications (up to 1% scaling) significantly alter pressure drop in the packed bed, with optimal fidelity achieved at a slight overlap ($\Delta D_p \approx 0.5\%$). Two turbulence models ($k-\omega$ Shear Stress Transport (SST) and $k-kl-\omega$) are assessed, revealing model-dependent sensitivities to contact treatments: the $k-kl-\omega$ model excels for small gaps ($-0.2\% \leq \Delta D_p \leq 0\%$), while SST performs better for large overlaps ($+0.2\% \leq \Delta D_p \leq +2\%$). A square domain that was 260 mm long and had a cross-section of 20×20 mm was packed with spherical particles in 2×2 stacked layers. The particle size ranged from 9.9 to 10.1 mm. Derived inlet velocity ranged from 3.7 to 9.5 m/s. Validation against experimental data shows a mean absolute error of $< 4.5\%$ for pressure drop predictions. The results provide quantitative guidelines for selecting contact point treatments in computational fluid dynamics simulations, balancing accuracy with computational feasibility. This work advances the modeling of packed beds by linking microscale geometric adjustments to macroscopic flow predictions, offering practical insights for reactor design and optimization.

1. Introduction

Packed bed reactors represent one of the most widely utilized configurations in engineering applications, finding extensive use in catalytic processes, absorption columns, and thermal energy storage systems [1,2]. The complex interplay between fluid dynamics and heat and mass transfer phenomena in these systems has long been a subject of intensive research. Computational Fluid Dynamics (CFD) has emerged as a powerful tool for investigating these intricate flow patterns, offering insights that are often difficult to obtain through experimental measurements alone [3–5]. However, despite significant advancements in computational capabilities, the accurate numerical representation of packed bed geometries remains a formidable challenge, particularly concerning the treatment of inter-particle contact points and their impact on fluid dynamic behavior [6,7].

The fundamental challenge in CFD modeling of packed beds stems from the inherent geometric complexity of randomly arranged and structurally ordered packed beds. For example, when spherical particles

are packed together, they form distinct contact points that create microscale flow constrictions, significantly influencing the macroscopic pressure drop and flow distribution [8–10]. These contact regions represent areas of particular interest, as they often play an important role in the overall fluid dynamic resistance. Traditional approaches to modeling packed beds frequently employ simplifications such as particle overlap or contact point elimination to facilitate mesh generation and numerical solution stability [11–13]. While these simplifications make the computational problem more tractable, they inevitably introduce errors in the prediction of key parameters, most notably the pressure drop across the bed.

Bu et al. evaluated contact point modification methods — gaps, overlaps, bridges, and caps — for CFD analysis of forced convective heat transfer in structured packed beds of spheres [14]. Using the RANS approach, the authors demonstrate that the bridges' method yields reasonable local heat transfer predictions across simple cubic, body-centered cubic, and face-centered cubic packings. While contact

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Nomenclature**Greek Symbols**

ϵ	Turbulent dissipation rate
ϵ	Void fraction (porosity) of the packed bed
μ	Fluid viscosity
ρ	Fluid density
v_s	Superficial velocity

Roman Symbols

ΔP	Pressure drop
ΔD_p	Particle diameter deviation
Δx	Mesh size
D_p	Equivalent spherical diameter of the particle
G_m	Fluid mass flow rate
I	Turbulent intensity
k	Turbulent kinetic energy
L	Length of the packed bed
L_m	Turbulent length scale
R_t	Turbulent viscosity ratio
y^+	Dimensionless wall distance

treatments significantly influence pressure drop predictions. Notably, the overlaps and bridges methods preserve realistic local flow near contact zones, whereas gaps and caps distort these profiles. For the gaps and overlaps modification methods, the authors employed particle diameter adjustments of $\pm 1\%$, using a 1% scale-down for gaps and a 1% scale-up for overlaps.

Pichler et al. investigated contact point treatment methods for CFD heat transfer simulations in DEM-generated packed beds containing up to 619 particles [15]. The study implemented $\pm 1\%$ particle diameter modifications — using 1% scale-down for gaps and 1% scale-up for overlaps — while proposing a novel local bridges method with corrected effective thermal conductivity. Their work provides critical insights into optimizing contact point treatments for accurate prediction of both heat transfer and fluid dynamics in resolved packed bed simulations.

Dixon et al. conducted a study of contact point treatments for 3D CFD simulations of fixed beds, examining particle–particle and wall–particle interactions at industrial-scale Reynolds numbers (500–10,000) [16]. The authors evaluated global modification methods using particles scaled to 0.97–0.99 (gaps) and 1.01–1.03 (overlaps) of nominal size, finding these uniform adjustments altered bed porosity and produced significant errors in drag coefficient predictions. Their analysis demonstrated that local methods — particularly bridge insertions at contact points — provided superior accuracy for both hydrodynamic (drag coefficient) and thermal (heat flow) predictions compared to gap or overlap approaches. The study established quantitative guidelines for estimating errors associated with different contact point treatments, concluding that bridge methods offer the best compromise for coupled flow and heat transfer simulations in packed beds.

Scale-up and scale-down approaches, even for changes as small as 1–3 lead to significant alterations in packed bed structure because these adjustments affect the entire computational domain — not just the contact point region — thereby either decreasing or increasing the packed bed porosity. According to the widely used Ergun equation for pressure drop prediction [17], the pressure drop is a function of packed bed porosity and particle size. Thus, even a minor change in particle size will alter the packed bed porosity and, consequently, the pressure

drop:

$$\Delta P = \frac{150\mu L}{D_p^2} \frac{(1 - \epsilon_p)^2}{\epsilon_p^3} v_s + \frac{1.75 L \rho}{D_p} \frac{(1 - \epsilon_p)}{\epsilon_p^3} v_s |v_s|, \quad (1)$$

where ΔP is the pressure drop across the packed bed, L is the length of the packed bed, D_p is the equivalent spherical diameter of the particle, ρ is the density of fluid, μ is the dynamic viscosity of the fluid, v_s is the superficial velocity, ϵ_p is the void fraction (porosity) of the packed bed.

It should be noted that while there are critical discussions regarding the use of the Ergun equation for packed beds [18,19], many authors have confirmed that, in general, pressure drop depends on particle size and packed bed porosity [20,21]. Thus, despite its non-universal coefficients, Eq. (1) successfully captures the fundamental factors that govern the flow.

Below are some studies on different approaches on the packed bed generation, with diameter deviation denoted as ΔD_p :

- Nijmeisland and Dixon — gaps only ($\Delta D_p = \{-0.5, -1, -3, -5\}\%$). The best result at -0.5% ; no correlation shown [22].
- Romkes et al. — gaps only ($\Delta D_p = \{-1, -2, -3, -5\}\%$). The best result is at -1% ; no correlation is shown [23].
- Calis et al. — gaps only ($\Delta D_p = \{-1, -2\}\%$). The best result is at -1% [24].
- Kutscherauer et al. — gaps only ($\Delta D_p = -2.5\%$) [25].
- Moghaddam et al. — gaps only ($\Delta D_p = -0.5\%$) [26].
- Guardo et al. — overlaps only ($\Delta D_p = 1\%$) [27].
- Bu et al. — gaps and overlaps ($\Delta D_p = \{-1, 1\}\%$) also bridges and caps [14].

Many CFD studies of packed beds with spherical particles address the issue of topological singularities at the contact points. The most straightforward ways are the global ones (creating artificial gaps or overlaps by scaling the particles uniformly). Some authors assume a fixed scaling factor (typically 0.5–2.5%) without clear clarification [25–27], whereas others determine the best scaling factor during preliminary studies [14,23,24]. However, none of them reveal any analysis of their findings.

The present study addresses a knowledge gap in contact point research by conducting a comprehensive investigation into the effects of particle scaling on CFD simulation results for packed bed systems. Unlike previous works that have examined only coarse scaling factors of $\pm(0.5 - 5)\%$ without detailed clarification, this work focuses on a finely resolved range of scaling parameters, examining both scaled-up and scaled-down (1.002 to 1.05 times original diameter) particle configurations. This systematic approach allows for the precise quantification of how small changes in particle geometry propagate through to macroscopic flow characteristics. The primary objective is to establish quantitative relationships between particle scaling factors and the resulting errors in pressure drop prediction, thereby providing researchers with concrete data to inform modeling decisions.

The investigation employs a rigorous methodology combining advanced geometric modeling and meshing techniques with high-fidelity CFD simulations. The packed bed geometries are generated using discrete element methods to ensure physically realistic particle arrangements, followed by precise scaling operations to create the test cases. The CFD simulations utilize state-of-the-art meshing approaches capable of resolving the intricate flow paths around contact points while maintaining computational efficiency through parallel processing strategies. Particular attention is given to the treatment of near-wall regions, where the combination of particle scaling and wall effects creates particularly challenging flow conditions.

Through this systematic investigation, the study aims to provide researchers and engineers with quantitative data and practical recommendations that will improve the accuracy and reliability of packed

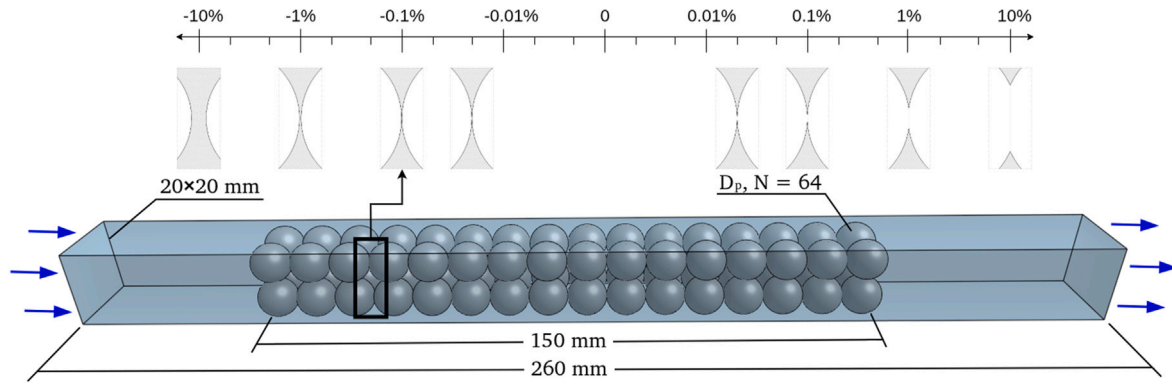


Fig. 1. Computational domain for CFD simulations and schematic illustration of gap and overlap treatments in packed beds.

bed simulations while maintaining computational feasibility. The results will help bridge the gap between idealized computational models and real-world packed bed systems, ultimately leading to better-designed industrial processes and more efficient utilization of energy and materials.

2. Methods

This paper examines the impact of inter-particle contact points on pressure drop in fluid flow through fixed beds using numerical simulations. Multiple bed configurations were analyzed by solving the Reynolds-averaged Navier–Stokes (RANS) equations, with experimental data serving as validation benchmarks for the computational results.

2.1. Computational domain and meshes

A square domain that was 260 mm long and had a cross-section of 20×20 mm was packed with spherical particles in 2×2 stacked layers (Fig. 1). Sixty-four spheres of diameter $D_p = 10$ mm each were arranged in an orthogonal array with a pitch of 10 mm. The entrance and the exit regions were about 50 mm each.

The centers of the spherical particles were fixed at nodes of an orthogonal grid, whereas their diameter varied in a range $\pm 10\%$ (Fig. 2). This variation produced required gaps and overlaps, ranging from $\pm 2 \mu\text{m}$ to ± 1 mm. The diameter deviation will be denoted as ΔD_p and expressed in percent, whereas gap/overlap size will be denoted as δ and given in absolute values (mm).

The computational domain was discretized using an unstructured mesh comprised of nonuniform polyhedra cells with a growth rate of 1.2 (Fig. 2). Up to five prismatic layers with a growth rate of 1.5 were applied to all fluid–solid interfaces to enhance solution versatility in high gradient regions. First layer thickness, growth rate, and number of layers varied across the mesh to ensure wall distance $y^+ \sim 1$ and minimize prismatic layers' stepping. Dimensionless distance postprocessing revealed actual y^+ values averaged around 4.2 with localized peaks up to 23. However, numerical residuals were stable, indicating that the wall functions successfully resolved the near-wall flow despite this local variability.

All meshes had an average cell count of $1.6 \cdot 10^6$, with variation less than 10%. Mesh quality was assessed in terms of orthogonality and skewness. The average aspect ratios of cell faces showed little variation (1.4 ± 0.37) and no noticeable trends. With the same discretization algorithm, both quality metrics had a distinct correlation with diameter deviation ΔD_p (Fig. 3). This phenomenon can be explained with an inflation algorithm that is used to generate prismatic layers near the walls. As the gap narrows, these layers distort cells between them (see $\Delta D_p = -4\%$) resulting in the first noticeable quality reduction (Fig. 3). Further convergence leads to prismatic layers stepping, as they cannot be applied near the prospective contact region (see $\Delta D_p = -2\%$). With a

positive deviation of the particles' diameter, the gap is eliminated due to overlap (see the right column in Fig. 2). Larger overlaps produce wider angles that are easier for the inflation algorithm to process. At some point, prismatic layer stepping is no longer required (see $\Delta D_p = 4\%$). However, mesh quality declines with larger values of ΔD_p , especially without stepping of prismatic layers. That is because of the rapidly increasing count of wedge-shaped cells with high aspect ratios near the overlap regions (see $\Delta D_p = 4\%$).

From meshes overview (Fig. 2) one can clearly distinguish three groups of meshes regarding prismatic layers: with smooth layers ($\Delta D_p \leq -4\%$), with stepped layers ($-4\% < \Delta D_p < 4\%$), and with converged layers ($\Delta D_p \geq 4\%$). These groups were expected to have a significant step-like effect in the mesh quality diagram and simulation results. However, this was not observed due to the implementation of wall functions in the turbulence models.

Maximum cell size Δx varied from 0.1 to 0.4 mm with a local refinement factor up to 2.5 (i.e., 0.04...0.16 mm). For the considered case, Richardson's extrapolations predicted pressure losses of 23.52 and 24.21 kPa (on theoretical mesh with element sizes $\Delta x \rightarrow 0$) for $k-\omega$ SST and $k-kl-\omega$ models correspondingly. With an acceptable error of 5% (green zone on Fig. 4), acceptable results were obtained on meshes with cell sizes up to 0.25 mm. Thus $\Delta x = 0.25$ mm was selected as a default mesh size for the main simulations.

2.2. Governing equations

The following basic formulation of Navier–Stokes equations was used:

$$\nabla \cdot \mathbf{u} = 0; \quad (2)$$

$$(\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu (\nabla^2 \mathbf{u}) + \nabla R_{ij}; \quad (3)$$

$$R_{ij} = \nu_t (\nabla^2 \mathbf{u}) - \frac{2}{3} k \delta_{ij}. \quad (4)$$

Velocity divergence in continuity equation (2) was eliminated due to the incompressibility of the working fluid (air). Momentum equation (3) has no time derivative due to steady-state formulation. There is also no gravity term, since the system was considered isothermal. The left-hand side of the momentum equation describes the spatial distribution of the velocity field, whereas the right-hand side contains forces causing the flow: a negative pressure gradient that drives the flow and viscous forces that smooth out velocity gradients and resist the motion.

Two turbulent models were used to estimate the Reynolds stress (eq. (4)): the shear stress transport model ($k-\omega$ SST proposed by F. Mentor in the early 1990s [28]) and the $k-kl-\omega$ model proposed by D. K. Walters and C. Davor in the late 2000s [29]. The former solves two additional transport equations to estimate kinetic energy k and its specific dissipation rate ω . Then turbulent viscosity ν_t can be estimated with an empirical correlation between k and ω . The latter model solves

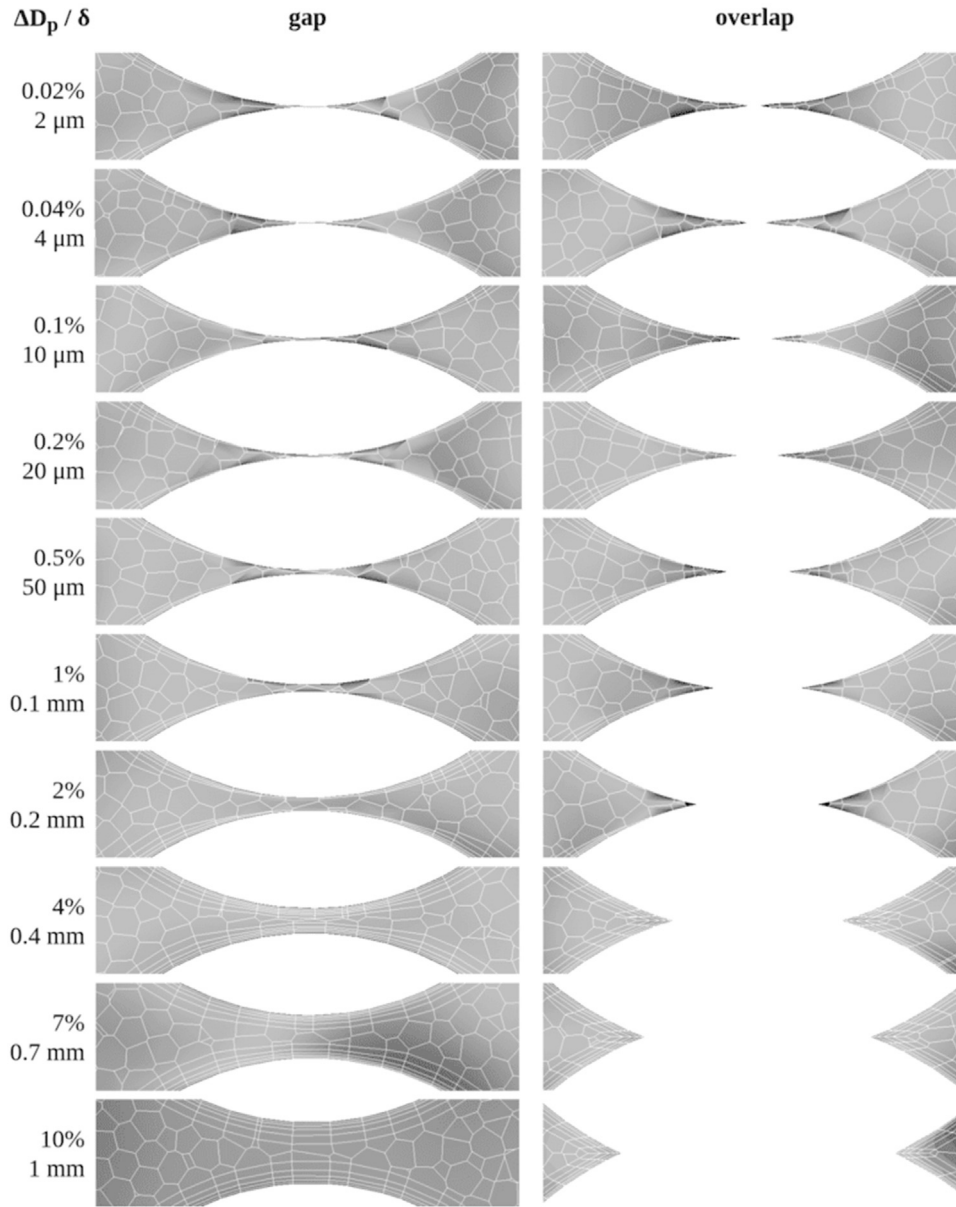


Fig. 2. Computational meshes overview with ΔD_p values.

three additional equations, providing predictions similar to those of SST for k and ω , but it assumes that the kinetic energy consists of both turbulent and laminar components: k_T and k_L . Similarly, the $k - k_l - \omega$ model operates with total turbulent viscosity that comprises large-scale and small-scale components.

The term $-\frac{2}{3}k\delta_{ij}$ in eq. (4) utilizes Kronecker delta to ensure the trace of the Reynolds stress tensor R_{ij} is zero for incompressible flows, separating volumetric effects from shear effects. The turbulent viscosity ν_t is used to account for an intensification of molecular diffusion by turbulent eddies; thus, the effective viscosity is $\nu_{eff} = \nu + \nu_t$.

2.2.1. Boundary conditions

At the inlet, a uniform velocity profile was set with the Dirichlet boundary condition: $u(x) = U_0 \vec{n}$. A flat velocity profile was used instead of a fully developed one because the contribution of wall shear stress (at entrance and exit regions) to the overall pressure drop is negligible compared to that of the packed bed. Velocity was derived from mass flow rates available in experimental data [30]. Turbulence parameters of the inlet flow were set with turbulent intensity $I = \frac{5.5}{Re^{0.0407}}$ and

turbulent viscosity ratio $R_t = \frac{0.09\rho k^2}{\epsilon\mu}$, where Re is Reynolds number, ρ and μ are fluid density and viscosity, $k = 1.5(U_0 I)^2$ is turbulent kinetic energy, and $\epsilon = \frac{0.164k^{1.5}}{L_m}$ is turbulent dissipation rate. The turbulent length scale $L_m = 0.4 \frac{0.382x}{Re_x^{0.2}}$ was calculated with the coordinate-related Reynolds number $Re_x = \frac{U_0 x \rho}{\mu}$, where x is the entrance region length.

At the outlet, a surface-normal zero gradient of pressure ($\partial p / \partial \vec{n}$) with $p_{gauge} = 0$ was set due to flow incompressibility. Additionally, the backflow was restricted to stabilize the solution, which was dynamically controlled using Eq. (5). Fluid–solid interfaces (i.e., channel and particle walls) were set with no-slip boundary conditions with artificial roughness of 45 μm for the square domain. As particles were polished in experiments, artificial roughness was not applied to them.

2.2.2. Fluid properties

Assumption of incompressible and isothermal flow reduced operational properties of the working fluids to density (1.194 kg/m³) and viscosity (1.81 · 10^{−5} Pa s).

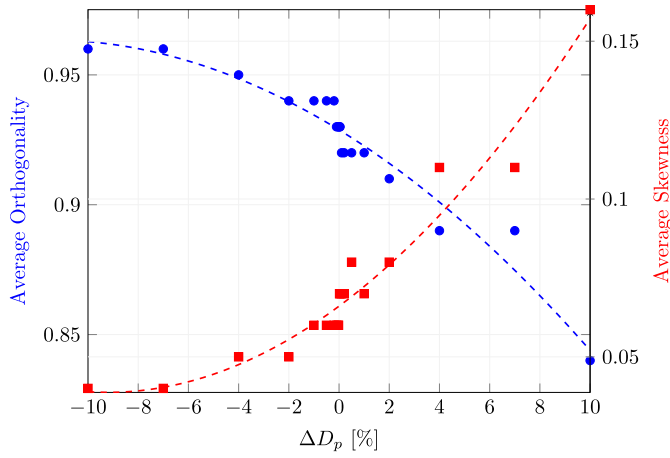


Fig. 3. Mesh quality assessment for different contact-point treatments (gaps and overlaps) in packed bed simulations: the relationship between particle diameter adjustments (ΔD_p) and mesh orthogonality/skewness metrics.

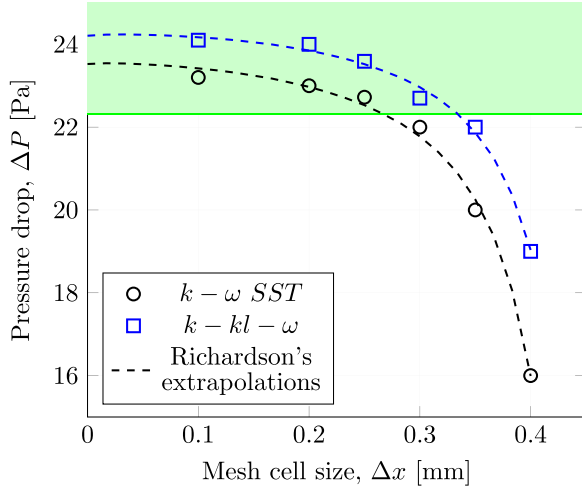


Fig. 4. Mesh dependency analysis.

2.3. Error analysis

To ensure an accurate and complete solution with the numerical model, convergence checks were applied during each simulation run. Control parameter (pressure drop) stabilization was continuously monitored over a moving average of 10 iterations with a stopping criterion of 0.1%:

$$\left| \sum_{i=10}^i \frac{p_{in} - p_{out}}{10} - \sum_{i=11}^{i-1} \frac{p_m - p_{out}}{10} \right| < 0.001(p_{in} - p_{out}). \quad (5)$$

Additionally, numerical residuals were continuously monitored with the following stopping criteria: $E_{continuity} = 0.01$, $E_{velocity} = 10^{-5}$, and $E_{pressure} = 10^{-6}$.

3. Results and discussion

3.1. Validation

Simulation results were in a good agreement with adopted experimental data [30] (Fig. 5). The presented in Fig. 5 results were obtained at a 4 μm gap between the particles ($\Delta D_p = -0.04\%$) as they had a minimum mean absolute error value. The average discrepancy over four data points was less than 0.55 kPa (2.2% of the full scale). Both

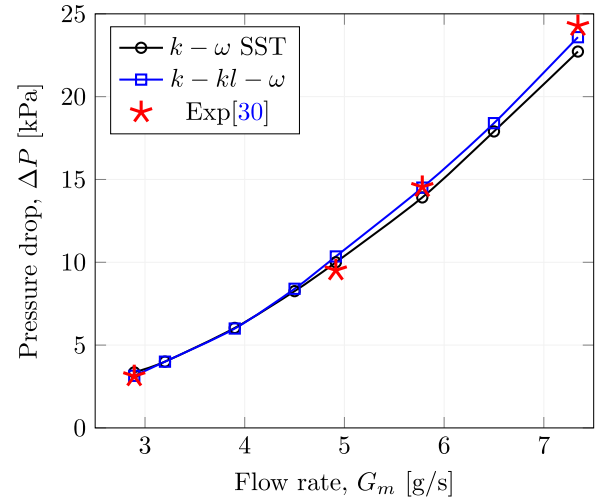


Fig. 5. Validation of CFD results against experimental pressure drop data [30].

models showed similar correlations between mass flow rate G_m and pressure drop ΔP . However, $k-kl-\omega$ model predictions were higher at large flow rates compared to $k-\omega$ SST results. This compensated for the major discrepancies at $G_m = 4.914$ g/s and $G_m = 7.342$ g/s.

3.2. Diameter deviation analysis

3.2.1. Flow analysis

The first phase of simulation results processing was dedicated to flow characterization. Although a turbulent flow has an intrinsically random nature, it is well known that chaotic behavior is usually microscale. This implies the global pattern of a developed flow should be rather stable in channels with smooth periodic topology. This also makes steady-state simulations with the Reynolds-averaged approach (RANS) an acceptable compromise between results fidelity and computational effort. Thus, a flow uniformity for the stated problem was considered a certain indicator of a well-behaved solution.

Simulation data was evaluated in terms of velocity and pressure distributions over the computational domain. Stream-wise distributions indicated flow development region length ranged from three to five calibers (i.e., 6–10 layers of particles or 60–100 mm) within the packed bed. Thus, transversal distribution was examined with offsets of 100 and 105 mm from the center of the first layer of particles (Fig. 6). The former cross-section goes through the spheres' centers (i.e., the narrowest part of the channel), whereas the latter one is in between (i.e., the widest part of the channel).

Cross-section at $x = 100$ mm contains the main center pathway, four half-sized pathways, and four quarter-sized pathways. These pathways are easily distinguishable when particles overlap. However, lateral overflow between the pathways was negligible compared to the stream-wise flow, even at large gaps. Thus, the 9-pathways structure held for the whole considered range of particle diameters.

The remaining ΔD_p configurations exhibit similar geometric features to those presented in Fig. 6. The minimum flow area occurs at the axial plane intersecting the sphere centers (narrowest constriction), while the maximum flow area appears at the midplane between adjacent particle centers (widest passage). These cross-sectional comparisons demonstrate the distinct flow channel morphologies resulting from different contact treatment approaches.

To visualize flow uniformity, the main pathway was selected as it transmitted fluid at the highest speed (Fig. 7). To compensate for an average velocity variation with ΔD_p , three sets of scales were used: 0–100, 0–150, and 0–200 m/s. The maximum flow rate of 7.342 g/s

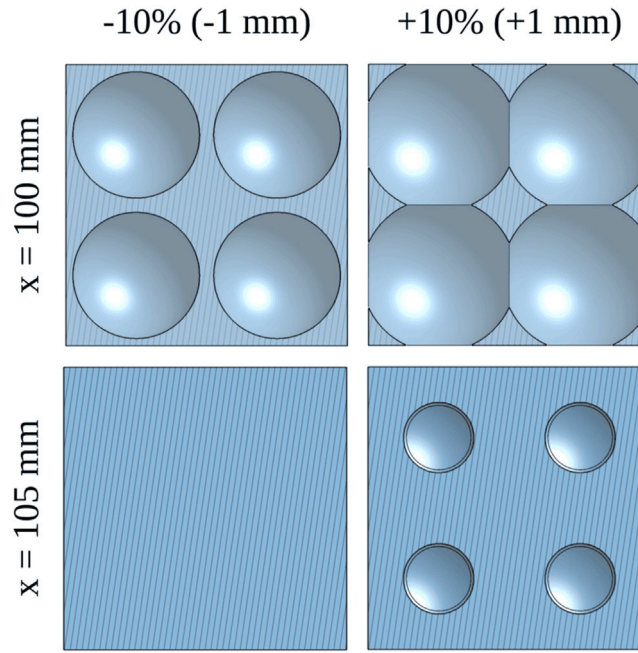


Fig. 6. Cross-sectional geometries of the packed bed at axial positions $x = 100$ mm (narrowest flow constriction) and $x = 105$ mm (widest flow area), comparing (left) gap ($\Delta D_p = -10\%$) and (right) overlap ($\Delta D_p = +10\%$) contact treatments: the flow channel morphology differences between configurations.

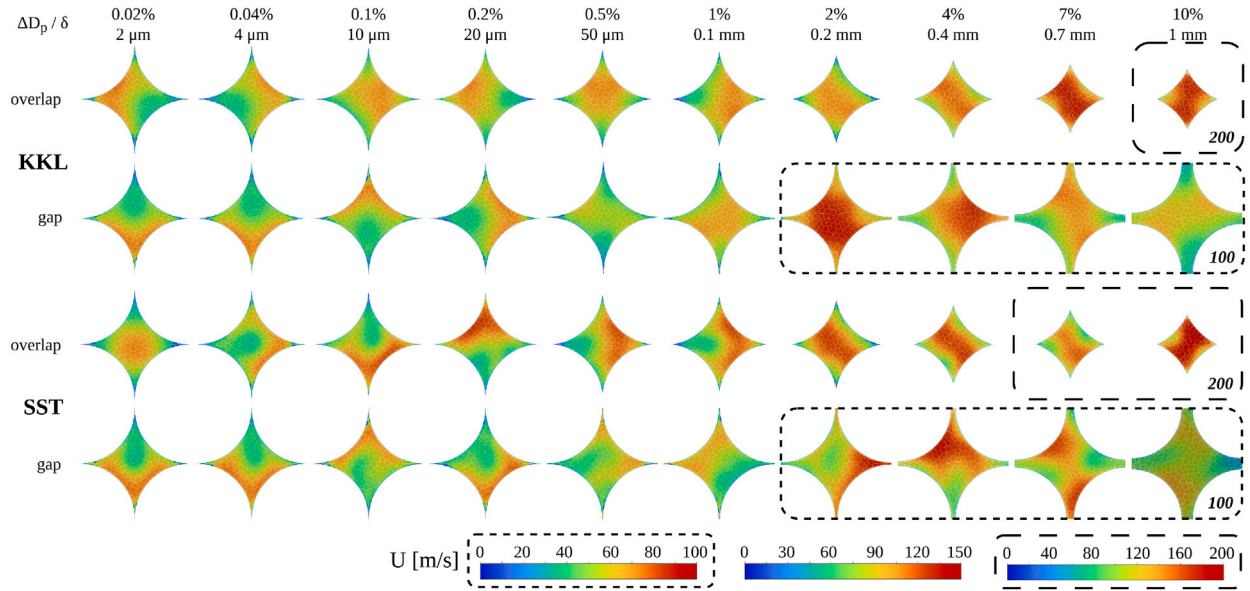


Fig. 7. Velocity contours at $x = 100$ mm and $G_m = 7.342$ g/s.

was used for visualization, as it provided the highest mean velocities that are the most challenging for simulation.

While the velocity distributions were mostly random, certain cases (e.g., the 2 μm overlap with the SST model) exhibited flow profiles similar to those in a uniform cross-section channel, despite the actual pathway having a varying lateral profile. This means in these cases the flow was more stable than elsewhere. One should note that this does not indicate lower fidelity in other cases because the average velocities were consistent. However, SST and $k-k_l-\omega$ models yielded noticeably different dispersions of velocity magnitudes in the mentioned cross-sections. With particles diameter deviation $\Delta D_p \geq 0.1\%$, the latter model yielded less dispersion (e.g., compare 10 μm overlaps). This indicated a more stable solution with the $k-k_l-\omega$ model.

Cross-sections at $x = 105$ mm showed the effect of pathways expansion on jets' shapes (Fig. 8). Regardless of the particles deviations, the ratio of hydraulic diameters at $x = 100$ mm and $x = 105$ mm stays the same. Thus, the flow imprints should have similar sharpness. However, that was clearly not the case. Some jets retained their original shape, whereas others exhibited deformation or breakdown along their pathways. One can see the smearing of the imprints in cases with gaps over 0.1 mm (note that the SST model yields more severe distortions). This implies both models (especially the SST) struggle with separation of the boundary layer that occurs over the spheres.

3.2.2. Pressure drop

Particles resizing with fixed allocation expectedly affects the hydraulic head of the packed bed. Growing particles ($\Delta D_p > 0$) increase

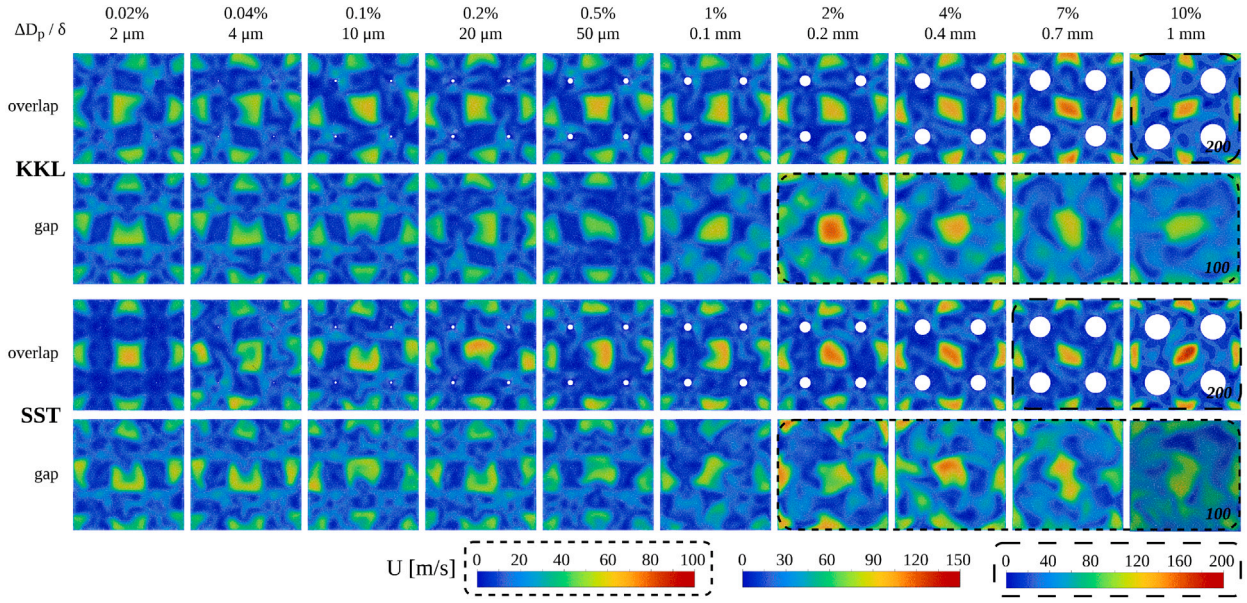


Fig. 8. Velocity contours at $x = 105$ mm and $G_m = 7.342$ g/s.

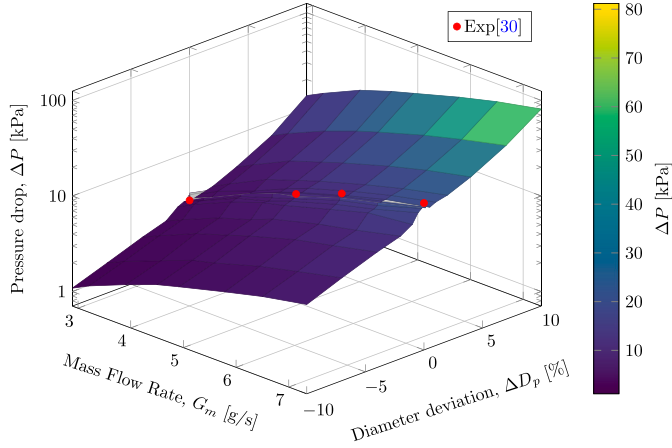


Fig. 9. Pressure drop versus mass flow rate for different contact-point treatments: (solid surface) numerical simulation results and (discrete points) experimental data from [30].

pressure drop due to shrinking of the pathways and vice versa. The correlation is similar to exponential regardless of the starting point (i.e., base flow rate or velocity) (Fig. 9).

A mean absolute error (MAE) was used to elaborate on the correlation between simulation fidelity and particles diameter deviation:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|. \quad (6)$$

Here, y_i and $\hat{y}_i$ are predicted (simulated) and expected (experimental) data points. This metric was preferred over the popular root mean squared error ($RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$) because MAE will not exaggerate large errors. That is critical in the considered case, as errors are not normalized.

The MAE analysis revealed that both the SST and $k-kl-\omega$ models produced comparable results for large negative deviations in particle diameter ($\Delta D_p \leq -1\%$) (Fig. 10). However, for moderate deviations ($-0.2\% < \Delta D_p < 0.5\%$), the $k-kl-\omega$ model demonstrated greater accuracy. In contrast, the SST model outperformed the $k-kl-\omega$ model for large positive deviations ($\Delta D_p \geq 1\%$).

Obtained correlations between ΔD_p and MAE have a distinct scoop-like shape, with the minimum at $\Delta D_p \approx 0.5\%$. This overall trend is most obvious in a regular scaled fill-range diagram (see top left plot in Fig. 10). This suggests that a slight overlap between particles yields better results. Moreover, the asymmetry in the diagram's flanks indicates that an overlap resulting from $\Delta D_p \leq 1\%$ produces outcomes comparable to those with a gap caused by $\Delta D_p \geq -0.2\%$. Consequently, this plateau is shifted towards positive deviations in particle diameter, making overlaps a safer choice in cases of nonuniform ΔD_p . However, outside this plateau, gaps are more favorable as the associated errors increase at a slower rate.

At a smaller scale, when diameter deviations are limited to $-0.5\% + 1\%$ range, fluctuations are obvious (see top right plot in Fig. 10). These can be averaged to a MAE = 6%. However, a more accurate solution (with MAE = 4.5%) can be obtained using gaps ($\Delta D_p \geq -0.2\%$) with the $k-kl-\omega$ model.

Results fluctuations around $\Delta D_p = 0$ might be caused by insufficient mesh resolution in prospective contact regions. In section views of the computational mesh (Fig. 2) one can see that small gaps ($\Delta D_p \geq -1\%$) were discretized with highly deformed cells. Small overlaps ($\Delta D_p \leq 1\%$) produce very narrow wedges that were poorly discretized as well. These regions could obviously be refined manually to produce high-quality mesh. However, the procedure requires either the development of a scripting routine or very tedious manual adjustments. Both were discarded intentionally because the study intended to highlight the challenges that might occur with a straightforward simulation approach. Note that the simulation successfully passed all typically recommended checks: mesh quality criteria and residuals convergence. Thus, the simulation results at any considered case of ΔD_p might be legitimately accepted as valid if not checked for continuity within a certain range of diameter deviation.

4. Conclusion

This study presents a systematic investigation of contact-point treatments in CFD simulations of packed bed flows, providing quantitative insights into their effects on pressure drop predictions. Through RANS-based simulations with two turbulence models ($k-\omega$ SST and $k-kl-\omega$), we demonstrate that minor particle diameter adjustments significantly influence flow morphology and pressure drop accuracy.

Three key findings emerge from this work. First, a slight overlap ($\Delta D_p \approx +0.5\%$) yields optimal fidelity, minimizing mean absolute error

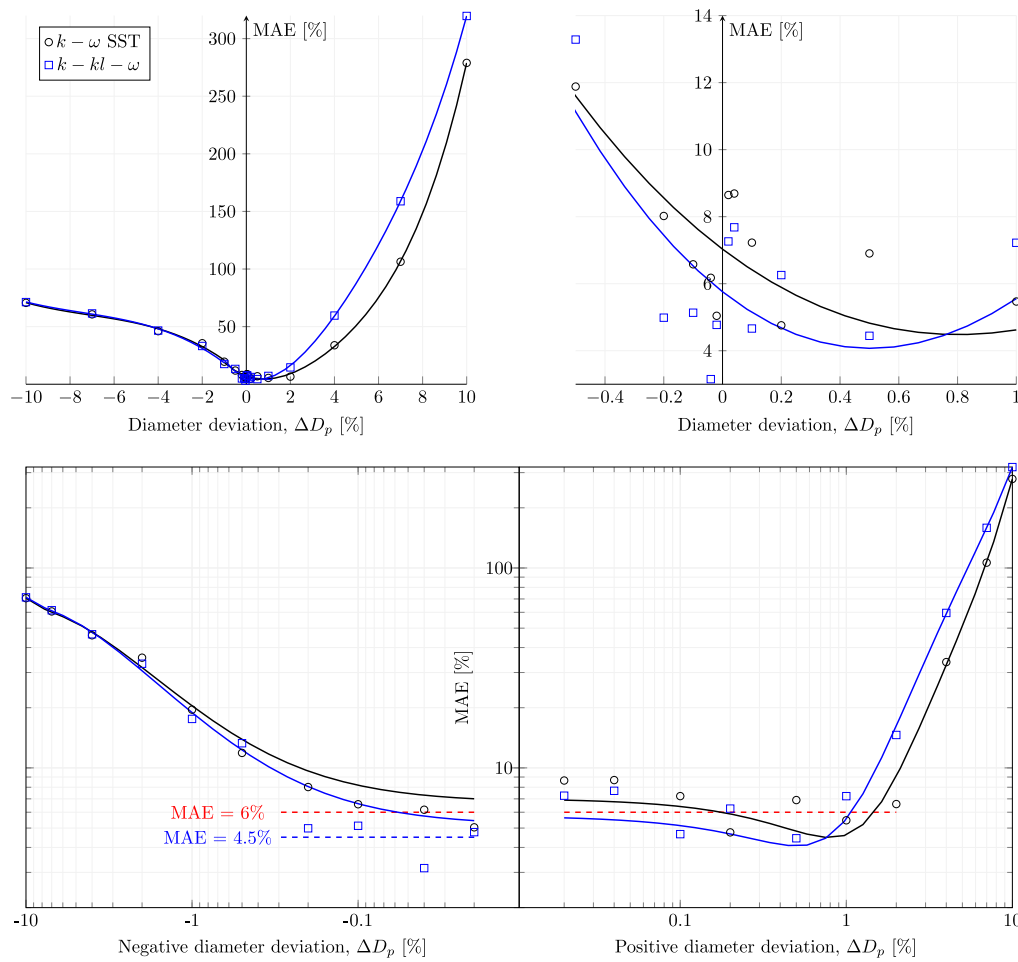


Fig. 10. Mean absolute error (MAE, Eq. (6)) analysis of pressure drop predictions for different contact-point treatments: hollow points — numerical results, solid lines — approximation curves, and dashed lines — expected MAE values for given design parameters.

(MAE < 6%) in pressure drop predictions compared to experimental data, with overlaps generally outperforming gaps for ΔD_p by up to +1%. Second, the k - kl - ω model shows superior accuracy for small gaps ($-0.2\% \leq \Delta D_p \leq 0\%$), while the SST model performs better for large overlaps ($+0.2\% \leq \Delta D_p \leq +2\%$), revealing important model-dependent sensitivities. Third, mesh quality analysis identifies critical trade-offs, with orthogonality and skewness degrading near contact points, particularly for small gaps and overlaps ($-2\% \leq \Delta D_p \leq +2\%$) where prismatic layer stepping becomes necessary. Cross-sectional flow analysis further demonstrates that gap treatments distort boundary layers more severely than overlaps, leading to jet smearing and higher velocity dispersion, especially when using the SST model.

These results advance packed bed modeling by establishing quantitative relationships between microscale geometric adjustments (ΔD_p) and macroscopic flow predictions. For industrial applications, we recommend a +0.5% overlap as the best compromise between accuracy and computational cost, with model selection guided by the dominant contact treatment approach (k - kl - ω for gap-dominated systems versus SST for overlap-dominated cases).

Future research should extend this work in several important directions. The framework developed here could be applied to non-spherical particles, including widely used shapes, e.g., cylinders and Raschig rings, to understand shape-dependent contact effects. The influence of particle roughness on pressure drop predictions and its interaction with contact treatments remains an open question that warrants detailed investigation.

This work bridges the gap between idealized CFD models and real-world packed beds, offering both immediate practical guidelines for reactor design and a clear roadmap for advancing the fundamental understanding of packed bed flows.

CRediT authorship contribution statement

Maxim Nikitin: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Dmitry Pashchenko:** Writing – review & editing, Writing – original draft, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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